

Custom Large Language Models Improve Accuracy: Comparing Retrieval Augmented Generation and Artificial Intelligence Agents to Noncustom Models for Evidence-Based Medicine



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Purpose: To show the value of custom methods, namely Retrieval Augmented Generation (RAG)-based Large Language Models (LLMs) and Agentic Augmentation, over standard LLMs in delivering accurate information using an anterior cruciate ligament (ACL) injury case. **Methods:** A set of 100 questions and answers based on the 2022 AAOS ACL guidelines were curated. Closed-source (open AI GPT4/GPT 3.5 and Anthropic's Claude3) and open-source models (Llama3 8b/70b and Mistral 8×7b) were asked questions in base form and again with AAOS guidelines embedded into a RAG system. The top-performing models were further augmented with artificial intelligence (AI) agents and reevaluated. Two fellowship-trained surgeons blindly evaluated the accuracy of the responses of each cohort. Recall-Oriented Understudy of Gisting Evaluation and Metric for Evaluation of Translation with Explicit Ordering scores were calculated to assess semantic similarity in the response. **Results:** All noncustom LLM models started below 60% accuracy. Applying RAG improved the accuracy of every model by an average 39.7%. The highest performing model with just RAG was Meta's open-source Llama3 70b (94%). The highest performing model with RAG and AI agents was Open AI's GPT4 (95%). **Conclusions:** RAG improved accuracy by an average of 39.7%, with the highest accuracy rate of 94% in the Meta Llama3 70b. Incorporating AI agents into a previously RAG-augmented LLM improved ChatGPT4 accuracy rate to 95%. Thus, Agentic and RAG augmented LLMs can be accurate liaisons of information, supporting our hypothesis. **Clinical Relevance:** Despite literature surrounding the use of LLM in medicine, there has been considerable and appropriate skepticism given the variably accurate response rates. This study establishes the groundwork to identify whether custom modifications to LLMs using RAG and agentic augmentation can better deliver accurate information in orthopaedic care. With this knowledge, online medical information commonly sought in popular LLMs, such as ChatGPT, can be standardized and provide relevant online medical information to better support shared decision making between surgeon and patient.

Shared decision making between a physician and patient is increasingly influenced by readily available research, clinical diagnoses, and treatments found

on the Internet. Having accessible information about a particular injury or condition and the associated treatments and outcomes can improve a patient's understanding of clinical options, perception of associated risks, and overall participation in the decision-making process of their own care plan.¹ Liira found that providing patients with accessible, accurate information about their health condition significantly improves both their understanding and their overall health outcomes in primary care settings.² Within orthopaedics, nearly two-thirds of patients utilize the Internet to obtain health and orthopaedic information.³ However, challenges associated with patient communication and expectation management persist. For example, patients may face difficulty discerning credible sources online,

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Received July 21, 2024; accepted October 27, 2024.

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0749-8063/241131/\$36.00

<https://doi.org/10.1016/j.arthro.2024.10.042>

which may lead to misleading information about a specific condition and treatment options. Furthermore, many resources utilize medical jargon, which is not well understood to those without a medical background. To mitigate these challenges, there exists a variety of resources aimed at using proper communication to present accurate and accessible evidence and information to support a well-informed decision. Such examples include reputable health websites and patient education materials dispensed by physician offices, which improve the integrity of the information provided.⁴ Similarly, improved readability through reduced complexity of written materials, and interactive visual aids, including videos, illustrations, and charts, simplify the information presented, thus improving comprehension levels.^{4,5} Despite this, research evidence and clinical values are rapidly evolving, and there remains a need for guidelines that are providing consistent, clinically valid, and current information to patients.

In this context, Large Language Models (LLMs) present a promising avenue. LLMs have demonstrated a broad range of exciting capabilities in generating text-based responses from immense volumes of data. Although the full extent of clinical application is still developing, these pretrained artificial intelligence (AI) systems can be used for knowledge retrieval, summarizing key research conclusions, and aid in clinical decision-making and communication. These models can also be accessed at all times by patients, alleviating the dependence on physicians for routine inquiries and addressing limitations in physician availability. Since the November 2022 release of OpenAI's popularized LLM, ChatGPT, other models have been produced and publicly provided at unprecedented speed, such as PaLM and Bard by Google AI, and Llama by Meta AI. To evaluate their potential in medicine in terms of response factuality, helpfulness, and possible harm, LLMs have recently been used to encode clinical information through the curation of broad medical question-answering (QA) benchmarks.^{6,7} LLMs have achieved state-of-the-art performance in US Medical Licensing Examination (USMLE) style questions on MedQA, with similar accuracy on other medical QA tasks such as MedMCQA, PubMedQA, and MMLU clinical topics datasets.⁸ LLMs have already had many various applications in the medical domain of orthopaedics specifically, such as for education, surgery, and research.^{9,10} While extremely powerful, an inherent shortcoming of LLMs is that the text generations stem from the unregulated domain of the Internet. Thus, these outputs, although generally agreeable, are at risk of being outdated, incorrect, generic, and unpredictable.¹¹ The content on the Internet has not and will unlikely ever be meticulously regulated.

However, like the Internet, there exists the possibility of establishing authoritative, truthful, and

evidence-based LLMs. One promising solution to optimize and improve LLM output involves Retrieval-Augmented Generation (RAG) technology and AI agents.¹² RAG takes the same impressive capabilities of LLMs and redirects this to retrieve relevant information from a specific domain of data. With RAG, the LLM can reference and cite information from authoritative, predetermined knowledge sources, rather than the nonspecific Internet. This predetermined knowledge refers to a corpus of text that the developer would want the LLM to reference and focus its responses around. In that sense, it would be manually curated, since the developer needs to specify and organize the data. This is akin to handing a student a textbook from which the questions on a test will be derived. Agents, on the other hand, are a team of multiple LLMs that each have a given role. To extend the analogy, an agentic system is like a student's group project, whereby each student is functioning within a particular assigned scope (i.e., researching, design, or quality assurance). In the same way that a well-coordinated group project can increase efficiency and quality over a single student's efforts, agentic systems offer opportunities for increasing the potential applications of LLMs. In our study, AI agents are additional LLMs that are assigned specific tasks based on the information retrieved by the RAG system to determine whether a response was appropriate. Between RAG and AI Agents, LLM responses are now primed to be more relevant, accurate, and reliable applications across many specific contexts when provided with an authoritative database or "textbook" from which to reference.^{13,14} Specifically in orthopaedics, management of anterior cruciate ligament (ACL) injuries involve layers of complexity related to timing, rehabilitation, and surgical versus nonsurgical indication. The rapid dissemination of evidence and new clinical guidelines makes this a prime use-case for AI integration. To show the value of custom methods, namely RAG-based LLMs and agentic augmentation, over standard LLMs in delivering accurate information using an anterior cruciate ligament (ACL) injury case. We hypothesize that including RAG and agentic adaptations to LLM will improve LLM accuracy rate.

Methods

Data Curation

The investigation was performed at the Long Beach Orthopedic Institute, Long Beach, California, and was exempt from Institutional Review Board approval. A set of 100 questions and answers based on the 2022 AAOS Anterior Cruciate Ligament (ACL) guidelines were curated by two fellowship-trained orthopaedic surgeons (P.N.R. and R.J.W.) (see [Appendix](#)). The primary components of the AAOS ACL guidelines included

summaries of recommendations/options (4 pages), recommendation/options (23 pages), and methodology (6 pages). To ensure that the entirety of the document was covered, 10 question-answer pairs were made for the respective summaries of recommendations/options and methodologies, while the 80-remaining question-answer pairs were on the core recommendation/options. Question-answer pairs were developed by taking facts from a particular section, and then providing a corresponding query for that topic that a surgeon or patient might have. For example, if the guidelines specify that “no new evidence has been given regarding aspiration of knee after an acute ACL tear,” then the corresponding question could be framed “is there currently high-quality evidence for aspiration for an ACL injury?” Every question was only a single sentence, and no patient vignettes were used. Patient vignettes were excluded to reduce bias, focusing exclusively on evaluating factual retrieval capabilities and framing this as a one-shot problem in terms of complexity.

Model Selection and Parameters

In the context of LLMs, closed-source refers to proprietary software with code that is not publicly available and typically requires a license, whereas open-source models have openly accessible model weights, allowing developers to freely build and adapt upon the base model. Thus, there remains debate about the trade-offs of utilizing a free, adaptable open-source model, or paying for state-of-the-art models for a given use-case. Both closed-source (Open AI GPT4/GPT 3.5 and Anthropic Claude3) and open-source models (Meta Llama3 8b/70b and Mistral 8×7b) were selected for this study. These models were selected due to their prominence, as well as their high performance on various LLM benchmark leaderboards.^{15,16}

RAG Embedding and Parameters

The text from the ACL guidelines pdf was extracted using PyMuPDF (Python Software Foundation, Wilmington, DE) and embedded into numerical vector representations. These representations are stored within a “vector database” that essentially serves as a library for numerical values that an LLM’s query will be compared to, allowing for input language to be quantitatively compared to the ACL guideline language. A “chunking” technique was applied in this process, whereby the vectors were limited to 1,000 characters, with 20-character overlap. This process allows for increased speed and relevance of retrieval of the embedded document. Therefore, when the user’s question is inputted, the model converts the question text into a numerical vector and calculates a “similarity” score from the vector database, ensuring that information in the answer relates to the initial question.¹⁷

The temperature parameter indicates an LLM’s “creativity and randomness”, whereby 0 draws more strictly from the embedded text while 1 indicates a higher variability and uses wider language. All models were set to temperature 0.7 to provide a balance of coherence and truthfulness to document.¹⁸

In the fast-evolving fields of computer science and AI, we may refer to certain framework published in arXiv, as this platform is critical for rapid dissemination of influential methodologies to the developer community for field testing and iteration, often preceding or complementing peer-reviewed publication.

Agents

LLM agents are systems of multiple different LLMs, each with different roles (i.e., task delegator) or access to tools (i.e., RAG or web-browsing). The rationale behind using agents is that more complicated tasks can benefit from multistep decision-making processes. This offloads the complexity to many simpler LLMs with a narrower purview instead of asking an LLM to do many tasks at once. For the purposes of fielding questions on ACL guidelines, the open-source and closed-source models with the highest evaluation metrics were supplemented with their own agentic system. This means that an LLM was allowed to interact with those models’ outputs and judge the retrieved responses from those models. In these systems, a RAG LLM would first read a question and conduct a search of the vector database. Then, instead of being immediately returned as the answer to the user, the RAG response was put through a Quality Judge LLM that determined whether the output was a relevant and appropriate answer to the initial question. This model was provided with a set of multishot prompts, which show examples of relevant responses (i.e., examples where a response addresses the core question comprehensively) and examples of irrelevant responses (i.e., referencing an entirely different procedure or treatment). If the Quality Judge LLM determined an erroneous output, it would repeat the RAG search until a satisfactory response was met. A summary of the workflow is described in [Figure 1](#).

Evaluation

The accuracy was evaluated with respect to the truthfulness to the embedded ACL guidelines by the two surgeons’ consensus. The surgeons were blinded to the models. Each question was scored either “1”, indicating faithfulness to the document or “0” for missing critical information within the answer. In instances of disputes, a consensus was decided by whether the response could be “reasonably” supported from the document with appropriate levels of specificity for the guidelines—such as questions with more than one right answer—or whether there was “little to no support” in the document for the response, even if the

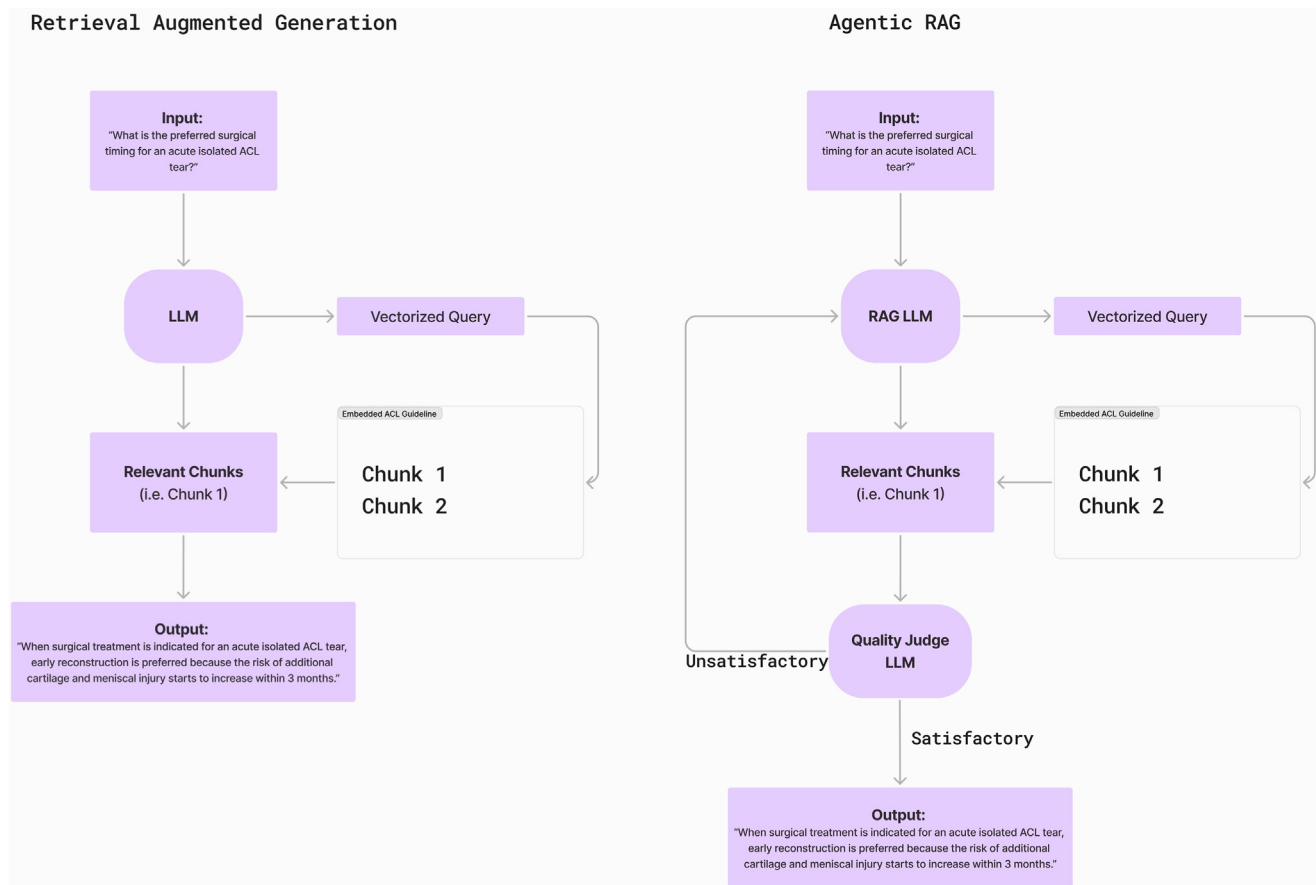


Fig. 1. Summary of the workflow where the Retrieval Augmented Generation–Large Language Models (RAG-LLM) would initially be prompted and then subjected to a Quality Judge LLM to determine the appropriateness of its response. If not found to be appropriate, the Quality Judge LLM would force the RAG-LLM to repeat the query until a satisfactory response to the original prompt was given.

statement itself was not wrong. The total number of correct responses was tallied for each model and reported as accurate. Additional common language-model metrics were also used. The Recall-Oriented Understudy of Gisting Evaluation (ROUGE) scores are commonly used for evaluating text summarization by measuring the overlap in words or longest shared subsequences between two bodies of text using n-grams, sequences of n words.¹⁹ For example, a 1-gram is a single word, and a 2-gram is a pair of words. ROUGE scores range from 0 to 1, with scores close to zero indicating poor similarity, and scores close to one indicating strong similarity. The threshold for a good score varies by n-gram type, with scores of 0.5 considered excellent for ROUGE-1 and 0.4 for ROUGE-2. While ROUGE scores are useful for evaluating certain aspects of language models, they do not assess the accuracy of the conveyed concepts or the grammatical correctness of the output. To capture more nuanced details of a response at the level of paraphrasing or synonyms, Metric for Evaluation of Translation with Explicit Ordering (METEOR) is often used for

comparing the alignment of generated text to references. This semantic detail is more flexible than the n-gram–based approach of ROUGE because it doesn't depend on precise overlaps and can help quantify levels of rewording. Much like ROUGE, METEOR scores range from 0 to 1, with a higher score reflecting superior translation quality.²⁰ Nevertheless, understanding highly niche responses related to ACL injuries cannot be evaluated by mere synonym or word-overlap metrics, and as a result of these innate limitations, the surgeon-rated accuracies remain the gold standard for evaluating the effectiveness of the information delivery by the LLMs. The source code for this evaluation can be found: https://github.com/jwoo153/ACL_RAG.

Results

All base models started <60% accuracy. Claude3 had the highest baseline accuracy of 58% compared to OpenAI GPT3.5 with the lowest baseline accuracy of 44%. Applying RAG improved the accuracy of every model, with an average improvement of 39.7%. The highest-performing RAG model was Meta's Open-

Table 1. Expert-Evaluated Accuracy, ROUGE1/2/L Scores, and METEOR Scores for Each Model in Its Base, RAG, and Agentic + RAG Setup

	Open-Source					
	Base			RAG		
	Accuracy (%)	ROUGE (1/2/L)	METEOR	Accuracy (%)	ROUGE (1/2/L)	METEOR
Meta						
Llama3 (70b)	54	0.37/0.16/0.29	0.35	94	0.54/0.42/0.50	0.63
Llama3 (8b)	57	0.12/0.03/0.09	0.18	94	0.46/0.34/0.42	0.54
Mistral						
Mistral8×7b	52	0.12/0.4/0.10	0.18	88	0.41/0.30/0.36	0.54
Closed-Source						
	Base			RAG		
	Accuracy (%)	ROUGE (1/2/L)	METEOR	Accuracy (%)	ROUGE (1/2/L)	METEOR
	Accuracy (%)	ROUGE (1/2/L)	METEOR	Accuracy (%)	ROUGE (1/2/L)	METEOR
OpenAI						
GPT4	49	0.20/0.07/0.15	0.25	93	0.50/0.39/0.46	0.66
GPT 3.5	44	0.18/0.06/0.14	0.26	90	0.53/0.40/0.47	0.60
Anthropic						
Claude3	58	0.17/0.07/0.13	0.25	93	0.31//0.24/0.28	0.44

NOTE. Agents were only incorporated into the RAG models with the highest evaluation metrics only—all other models were denoted with “N/A” since agentic systems built on less accurate models would likely compound underperformance. In the 1/2/L format, “1” indicates 1-word overlaps, “2” indicates 2-word overlaps, “L” represents the longest sequence overlap.

Source Llama3 70b (94%). However, GPT4 with RAG and AI agent augmentation had the highest overall accuracy of 95%. The average ROUGE 1/2/L score for the base models was 0.19/0.13/0.15 compared to the average RAG ROUGE 1/2/L score of 0.46/0.35/0.42, indicating a substantially greater lexical overlap in the words of the RAG-based output and the reference answer. The average METEOR score for base (0.25) was also lower across the boards compared to the RAG model average (0.57), suggesting increased semantic similarities between the reference answer and the generated answer. These findings are demonstrated in Table 1.

Discussion

Supplementing LLMs with RAG and agentic techniques showed improved performance in addressing document-specific questions related to the AAOS 2022 ACL guidelines when evaluated with respect to accuracy, improving by an average of 39%. These results demonstrate an advantage in reliability compared to non-custom LLMs, which draw their insight from the unregulated Internet. In spite of its limitations, LLMs embody a generative AI tool trained on a diverse wealth of information, including academic and medical knowledge that can and should be appropriately configured by domain experts to deliver timely and accurate information at the point of contact. Customization, as offered and demonstrated by the integration of RAG and agentic augmentation technology into LLMs, offers considerable promise in enhancing the accuracy and reliability of information, particularly in the field of medicine, where access is increasingly prioritized while combating misinformation. Using ACL injury information as a case example, our study highlights the potential of custom RAG- and agent-based LLMs to draw from information from particular source-text within the query responses, which is an important anchor for disseminating updated information in the midst of constantly changing practices. There were 3 main findings in our study. First, applying RAG improved the accuracy of every model by an average of 39.7%. Another important finding is that open- and closed-source models are comparable in performance. Lastly, incorporation of AI agents can improve RAG-LLM outputs and warrant more research. As new advancements continue to be made in custom LLM systems, the development of shared domain-specific evaluation sets and human feedback integration will be necessary for creating robust and generalizable standards. Data quality and collection will also become a foundational pillar of these systems, and the way that we structure information within guidelines or records must become increasingly optimized for model readability to ensure that we minimize the barriers to highly accurate models.

Several studies have explored the application of LLMs in medicine, highlighting their capacity to generate text-based responses from vast datasets.²⁰⁻²³ Notably, LLMs have demonstrated state-of-the-art performance on various medical QA benchmarks.^{9,10,23} These achievements have paved the way for LLMs to be utilized in medical education, surgical planning, and clinical research within orthopaedics as well. Kaarre et al. found that LLMs like ChatGPT could act as potential supplementary tools to further understand ACL surgeries in patient and nonorthopaedic medical doctor populations.²⁴ Truhn et al. used Chat-GPT4 to recommend treatment options based on MRI reports in patients with acute and chronic knee and shoulder conditions and found treatment options to be relatively appropriate.²⁵

While LLM has unlocked our collective potential to ask complex and multidimensional questions with seamless responses, there are still inherent limitations of generalized LLMs that must be addressed to optimize their clinical application. LLMs typically draw information from the unregulated domain of the Internet, which can lead to outdated, generic, and sometimes erroneous outputs.²⁶ The risk of these models regurgitating misinformation, “hallucinations”, and biases from nonauthoritative sources on the unregulated Internet are high. Dubin et al. highlighted this concern by comparing frequently asked questions and responses regarding total knee and hip arthroplasty between a Google Web search and ChatGPT, revealing heterogeneous and often inaccurate answers.⁷ To date, challenges remain in standardizing the quality of online medical information. Yang et al. previously compared LLM outputs from ChatGPT and Bard for treatment options to hip and knee osteoarthritis patients.²⁷ They found that both LLMs often recommended treatments that were not in line with current AAOS guidelines, highlighting the lack of reliability in LLM models currently. We found similarly poor and unreliable answers when evaluating base LLMs that were not augmented, with less than 60% accuracy. However, our study extends this experiment by examining questions beyond categorical recommendations. Furthermore, with the utilization of RAG techniques, we found that LLMs can draw source-specific information from the guidelines and, thus, answer guideline-based questions with higher accuracy.²⁸ Our study underscores the need for a nuanced discussion about the development of custom LLM workflows for specific use cases. Using ACL guidelines, we show how physicians may potentially benefit from accessing reliable information about the current clinical consensus for a given procedure. Kaarre et al. attempted to evaluate LLMs as a supplementary education with consideration to both patient and nonspecialist physician audiences, revealing an important and nuanced area of study regarding information delivery. We posit that

similar augmentation techniques in our study may potentially benefit patients’ education, who could interface with custom models to answer queries related to preoperative or postoperative instructions that are provided by their surgeon. More broadly, hospitals may also develop custom workflows to keep track of longitudinal patient-specific information in pursuit of more personalized care. Just as AAOS ACL guidelines can be accessed on the official website hosted by the AAOS, so can the custom AAOS-sponsored LLM—as well as its surgeons—on their own individual practice websites.

Our study found significant improvements in accuracy by an average of 39.7% when RAG technology was applied, with the highest performing RAG model, Meta’s open-source Llama3 70b, achieving a 94% accuracy rate. Examples of RAG use in medicine are slowly drawing peer-reviewed attention. For instance, a study on osteoarthritis management highlighted the limitations of generalized LLMs; after implementing RAG capabilities, without AI Agents, there was improved accuracy and relevance in the responses.²⁹ Another significant contribution is the Medical Information Retrieval-Augmented Generation Evaluation (MIRAGE), which compiled an evaluation benchmark and best practices for RAG specific to medicine from five medical QA datasets.³⁰ These examples illustrate the growing recognition of RAG’s potential to enhance the quality of medical information retrieval and generation.

While Meta’s open-source LLM had the highest accuracy rating after RAG-augmentation, overall accuracy rates in both open-(Meta Llama3 8b/70b and Mistral 8×7b) and closed (Open AI GPT4/GPT 3.5 and Anthropic Claude3) source models were relatively similar in our study. Open- and closed-source models primarily differ in their user accessibility. Open-source models allow individuals to understand and modify underlying algorithms to suit specific needs and locally host the LLM (i.e., privately and offline), while closed-source models are proprietary software pretrained by their initial developers.³¹ Although our study found both to be comparable initially, the intrinsic nature of LLM software can alter usage implications for large hospitals and medical organizations. From a privacy perspective, open-source LLMs offer the advantage of complete control over data handling. Medical institutions can host these models on their own secure servers, ensuring that sensitive patient information is not exposed to external entities. This level of control is crucial for maintaining patient confidentiality and complying with stringent health care privacy regulations, such as HIPAA in the United States. Additionally, reproducibility is another critical factor where open-source models have a distinct advantage. The transparency of open-source LLMs allows for reproducible research, as other researchers can access the same code

and datasets, verify results, and build upon previous work. This can potentially foster collaboration and accelerate advancements in medical AI. However, open-source LLMs often require experts to maintain and update these models, often improving with time, rather than providing robust capabilities from onset.³² Additionally, the computational resources needed to run large open-source models can be significant, potentially offsetting some of the initial costs saved from licensing fees.³² Thus, our study underscores the importance of future research in leveraging both open and closed source models to achieve optimal outcomes in medical information retrieval and generation.

Lastly, we found that incorporation of agentic systems can enhance RAG-LLM response accuracy, although these results seem to be model-dependent. Agentic augmentation further improves the accuracy of LLM outputs beyond what standard RAG can achieve, enhancing the model's capability to not only remain faithful to source text but also iteratively self-checking based on the outputs as well.³³ This can lead to more dynamic and responsive decision-making processes within the RAG framework, improving its overall effectiveness, as more agents are delegated roles, becoming more robust to real-world implications. Low et al. found that while RAG was able to improve the relevance of LLM responses, only agentic augmentation was able to answer novel questions compared to other LLMs in response to general health-related clinical questions.³⁴ Our study found that when GPT4 was augmented with an Agentic RAG system, it reached an overall accuracy of 95% from 93%, while the Llama3 70b dropped to 93% from 94%. This raises questions about the inherent predisposition of certain models to excel in agentic tasks. Previous literature has found that underlying architectural or training differences might influence the effectiveness of agentic modifications. A study by Brown et al. found that closed-source models, like GPT-4, outperformed open-source models in tasks requiring complex decision-making and context understanding due to their extensive training and optimization.³⁵ Sharma et al. demonstrated similar results, where closed-source models like GPT-3, and GPT-4 demonstrated significantly better agency, or the ability to self-automate correctly, than open-source models like Llama2 7b/13b/70b. They found that despite the potential for agentic automation, Llama2 variants were unable to effectively self-regulate or sustain their tasks.³⁶ These findings align with the variations between Llama3 70b and GPT-4 seen in our study, indicating that some models may possess intrinsic qualities that make them more amenable to agentic augmentation. It is possible that this phenomenon is explained by the agent "overthinking" the answer and making erroneous judgments of otherwise correct responses, suggesting improvement for internal agent optimization.

Future directions for this research should focus on expanding the application of RAG and agentic systems across various medical domains to identify their capabilities as information providers, task delegators, or diagnostic aids. Currently, further research should attempt to isolate the capabilities of LLMs at the level of prompt engineering, RAG, and agents when handling various levels of complexity, including multistep problems that simulate real-world situations. Additionally, human evaluators of various levels of expertise should be incorporated into future studies to further investigate the impact that LLMs have in conveying information to different audiences effectively. Further studies are needed to refine these technologies, explore their integration into clinical workflows, and assess their impact on patient outcomes. Additionally, developing comprehensive guidelines and best practices for implementing RAG in medical LLMs will be crucial for standardizing their use and ensuring their efficacy in delivering evidence-based, up-to-date information. Given the rapidly evolving nature of the medical field, future research may benefit from incorporating multiple guidelines and examining whether the latest guidelines are followed in agentic RAG-LLM outputs. Additionally, further work should be done to account for both the variabilities in question phrasing, as well as identification of knowledge gaps that are not within the embedded guidelines.

Noncustom LLMs have been met with intrigue but appropriate skepticism, given variably accurate response rates. When appropriately enhanced with domain-specific information and tools trained on an evidence-based foundation, custom LLMs can be exceptionally accurate and effective liaisons of information. RAG and agents embody critical technical LLM adaptations that offer physicians the ability to create evidence-based, accurate information for patients to maintain ownership in the dissemination of truth and reality in medicine.

Limitations

Our study is not without limitations. The accuracy of the models was evaluated by only two surgeons, which may introduce potential bias. Including multiple evaluators with diverse perspectives could help mitigate this and provide a more balanced assessment. Additionally, our study was constrained to a specific set of 100 questions derived from the 2022 AAOS ACL guidelines. Furthermore, the curated questions were single sentences and had corresponding existing answers in the guidelines. Real input data will likely be complex multisentence queries that may ask for information beyond the scope of the guidelines. Additionally, the questions did not contain any patient vignettes or ask to diagnose or plan treatments, and lack of testing in actual clinical scenarios make it difficult for further evaluation of the model to be applied to general applications.

Conclusions

RAG improved accuracy by an average of 39.7%, with the highest accuracy rate of 94% in the Meta Llama3 70b. Incorporating AI agents into a previously RAG-augmented LLM improved ChatGPT4 accuracy rate to 95%. Thus, agent- and RAG-augmented LLMs can be accurate liaisons of information, supporting our hypothesis.

Disclosures

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: P.N.R. has received support from Pacira for consulting, Stryker for consulting, Globus for consulting and royalties, Intelligent Health Analytics, Inc., for ownership and overture for ownership. The remaining authors (J.J.W., A.J.Y., R.J.O., S.S.H., D.H.N., B.U.N., R.J.W.) declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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Appendix: Question and Answer (QA) Script

Recommendations

Q1: What should a relevant history for assessing ACL injury include at minimum?

A1: A relevant history should include at minimum: mechanism and date of injury, history of hearing/feeling a popping sensation, ability to bear weight, ability to return to play, history of mechanical symptoms of locking or catching, localization of pain, if possible, and any history of prior knee injuries. (p. 20)

Q2: What should a physical exam for assessing ACL injury include at minimum?

A2: A physical exam should include at minimum: a neurovascular exam of the lower extremity with documentation of both distal perfusion and tibial/peroneal nerve function, assessment of varus and valgus laxity at 0 and 30° of flexion, dial testing at 30 and 90° of flexion, and evaluation of anterior-posterior laxity with Lachman's and anterior drawer and rotational laxity with pivot shift and active buckling sign tests.

Q3: When is early ACL reconstruction preferred over delayed reconstruction?

A3: Early ACL reconstruction is preferred when surgical treatment is indicated for an acute isolated ACL tear, because the risk of additional cartilage and meniscal injury starts to increase within 3 months.

Q4: Does early versus delayed ACL reconstruction affect range of motion outcomes?

A4: No, range of motion was not affected by early versus delayed surgery across multiple studies.

Q5: Is there a significant difference in outcomes between single- and double-bundle ACL reconstruction techniques?

A5: No, the majority of high-quality studies demonstrate no statistically significant difference in outcome parameters between single- and double-bundle ACL reconstruction techniques.

Q6: What do meta-analyses show regarding differences between single- and double-bundle reconstruction?

A6: Five meta-analyses demonstrated no statistically significant difference between single- and double-bundle reconstruction in postoperative pain, Lysholm, or IKDC subjective knee scores.

Q7: Which graft type is favored for ACL reconstruction in young and active patients?

A7: Autograft is favored over allograft for ACL reconstruction, particularly in young and/or active patients, to improve patient outcomes and decrease ACL graft failure rate.

Q8: What advantages does autograft have over allograft for ACL reconstruction?

A8: Autograft has advantages of lower graft ruptures/revisions, improved IKDC scores, better knee laxity

outcomes, and higher rates of return to activity compared to allograft.

Q9: What are the potential benefits of using bone-patellar tendon-bone autograft for ACL reconstruction?

A9: BTB autograft may reduce the risk of graft failure or infection compared to hamstring autograft.

Q10: What are the potential benefits of using hamstring autograft for ACL reconstruction?

A10: Hamstring autograft may reduce the risk of anterior or kneeling pain compared to BTB autograft.

Q11: Can training programs reduce the risk of primary ACL injuries in high-risk sports?

A11: Yes, training programs designed to prevent injury can be used to reduce the risk of primary ACL injuries in athletes participating in high-risk sports.

Q12: What types of exercises are typically included in ACL injury prevention training programs?

A12: ACL injury prevention training programs typically include strengthening maneuvers, plyometric exercises, and sport-specific agility drills.

Q13: When would ALL reconstruction or LET be considered during ACL reconstruction?

A13: ALL reconstruction or LET could be considered when performing hamstring autograft reconstruction in select patients to reduce graft failure and improve short-term function.

Q14: What are some drawbacks of adding ALL reconstruction or LET to ACL reconstruction?

A14: Potential drawbacks include additional incisions, implants, time under anesthesia, and unclear long-term outcomes, including a potential increased risk of lateral compartment osteoarthritis.

Q15: For ACL tears indicated for surgery, which treatment is recommended?

A15: ACL reconstruction is recommended over ACL repair because of the lower risk of revision surgery.

Q16: How do postoperative function outcomes compare between ACL reconstruction and repair?

A16: The evidence is mixed regarding postoperative function, with some high-quality studies favoring reconstruction and others favoring repair.

Q17: What is the guideline recommendation regarding aspiration of painful or tense effusions after knee injury?

A17: In the absence of reliable evidence, it is the opinion of the workgroup that physicians may consider aspirating painful, tense effusions after knee injury.

Q18: What are some potential benefits and risks of knee aspiration after acute ACL injury?

A18: Potential benefits include decreased pain, improved early range of motion, and possibly decreased presence of cytokines implicated in osteoarthritis progression. The main risk is the potential to introduce infection in an acutely injured knee.

Q19: Why might ACL reconstruction be considered, particularly in younger and/or more active patients?

A19: ACL reconstruction can be considered to lower the risk of future meniscus pathology or procedures, particularly in younger and/or more active patients.

Q20: How does ACL reconstruction compare to nonoperative treatment in terms of long-term pain and function?

A20: ACL reconstruction may be considered to improve long-term pain and function compared to nonoperative treatment, although evidence is limited.

Q21: What should be considered in patients with combined ACL tear and meniscal tear?

A21: In patients with ACL tear and meniscal tear, meniscal preservation should be considered to optimize joint health and function.

Q22: How does meniscal repair compare to meniscal resection in terms of future knee procedures?

A22: Meniscal repair has been associated with a higher rate of future knee procedures in the short term, particularly for medial meniscus repairs, compared to meniscal resection.

Q23: What is the recommended treatment for MCL in patients with combined ACL and MCL tears?

A23: In patients with combined ACL and MCL tears, nonoperative treatment of the MCL injury results in good patient outcomes, although operative treatment of the MCL may be considered in select cases.

Q24: When might operative treatment of the MCL be considered in combined ACL/MCL injuries?

A24: Operative treatment of the MCL might be considered in cases with greater MCL laxity or when there is a “floating meniscus” found during arthroscopy, as it may decrease the risk of subsequent ACL failure.

Q25: Is prophylactic knee bracing recommended to prevent ACL injury?

A25: No, prophylactic bracing is not a preferred option to prevent ACL injury.

Q26: What are some potential drawbacks of relying on prophylactic knee bracing for ACL injury prevention?

A26: Reliance on prophylactic bracing could decrease participation in injury prevention exercise programs, which have been shown to be protective against ACL and other lower extremity injuries. Additionally, biomechanical alterations from bracing may predispose to other injuries.

Q27: What type of functional evaluation may be considered as one factor to determine return to sport after ACL reconstruction?

A27: Functional evaluation, such as the hop test, may be considered as one factor to determine return to sport after ACL reconstruction.

Q28: Is there clear evidence regarding the optimal timing of functional testing and return to sport after ACL reconstruction?

A28: No, the optimal timing of functional testing and return to sport is unclear on the basis of the current literature.

Q29: Are functional knee braces recommended for routine use in patients who have received isolated primary ACL reconstruction?

A29: No, functional knee braces are not recommended for routine use in patients who have received isolated primary ACL reconstruction, as they confer no clinical benefit.

Q30: What are some limitations of the current evidence regarding functional bracing after ACL reconstruction?

A30: Limitations include the use of potentially inferior braces compared to modern custom options, relatively short-term follow-up, variability in bracing protocols, and relatively small study cohorts that may be underpowered for some outcomes.

Q31: Why is a history of hearing/feeling a popping sensation important when assessing for ACL injury?

A31: A history of hearing/feeling a popping sensation is important because it is often associated with ACL injury and can be a predictor of ACL tear.

Q32: What is the range of flexion dial testing that should be performed?

A32: Dial testing should be performed within 30–60° of flexion.

Q33: How soon after injury does the risk of additional cartilage and meniscal injury start to increase?

A33: The risk of additional cartilage and meniscal injury starts to increase within 3 months after the initial ACL injury.

Q34: Does early ACL reconstruction affect patient satisfaction compared to delayed reconstruction?

A34: There is no convincing evidence that early versus later reconstruction had an impact on general patient satisfaction.

Q35: Are there any differences in postoperative pain between single and double-bundle ACL reconstruction?

A35: No, meta-analyses have shown no statistically significant difference in postoperative pain between single- and double-bundle ACL reconstruction.

Q36: How do the costs compare between single- and double-bundle ACL reconstruction?

A36: Double-bundle ACL reconstructions involve increased surgical time and increased costs compared to single-bundle reconstructions.

Q37: Are there any potential advantages of using allograft for ACL reconstruction?

A37: While autograft is generally favored, allograft may have some advantages, such as reduced surgical time and absence of donor site morbidity.

Q38: How does the risk of disease transmission compare between autograft and allograft?

A38: Autograft eliminates the risk of disease transmission that exists, albeit low, with allograft use.

Q39: Which autograft source is associated with a lower rate of graft rupture/failure?

A39: Bone-patellar tendon-bone (BTB) autograft is associated with a lower rate of graft rupture/failure compared to hamstring autograft.

Q40: Are there any differences in revision rates between BTB and hamstring autografts?

A40: Yes, some studies have demonstrated lower revision rates with BTB autografts compared to hamstring autografts.

Q41: What specific outcomes have been shown to improve with ACL injury prevention training programs?

A41: Studies have shown reductions in the rates of complete ACL tears, total knee injuries, and overall injuries in athletes performing ACL injury prevention training programs.

Q42: Are ACL injury prevention training programs equally effective for male and female athletes?

A42: While effective for both, some moderate quality studies have shown particularly significant reductions in ACL injury rates for female athletes participating in these programs.

Q43: How does the addition of ALL reconstruction or LET affect the risk of graft failure in ACL reconstruction?

A43: Studies have reported a lower rate of graft rupture/failure when ALL reconstruction or LET is performed at the time of ACL reconstruction with a hamstring autograft.

Q44: Is there evidence regarding the long-term impact of ALL reconstruction or LET on knee osteoarthritis?

A44: Long-term impacts are unclear, but one study reported a higher rate of lateral compartment osteoarthritis in patients who underwent LET, although these patients also had a higher rate of partial lateral meniscectomy.

Q45: How do revision rates compare between ACL repair and ACL reconstruction?

A45: Studies show a lower rate of revision ACL surgery in patients undergoing primary reconstruction than in those undergoing repair.

Q46: Are there any potential advantages of ACL repair over reconstruction?

A46: Some studies have reported better postoperative function with ACL repair, but the evidence is mixed, and the higher revision rate is a significant concern.

Q47: Is there recent high-quality evidence regarding the benefit of knee aspiration after acute ACL injury?

A47: No, the guideline notes that no relevant articles have been published in the last 20 years regarding the benefit of aspiration in acutely injured knees.

Q48: How does knee aspiration potentially affect the progression to osteoarthritis following ACL tear?

A48: Acute aspiration may decrease the presence of cytokines, which may be implicated in the cascade progressing to osteoarthritis following ACL tear, but there is still no strong evidence.

Q49: How does ACL reconstruction compare to nonoperative treatment in terms of the risk of future meniscus injury?

A49: Some studies demonstrate a lower risk of future meniscus injury or secondary meniscectomy after ACL reconstruction compared to nonoperative treatment.

Q50: Is there a difference in return to activity rates between ACL reconstruction and nonoperative treatment?

A50: Current evidence shows no significant difference in return to activity based on available studies comparing ACL reconstruction to nonoperative treatment.

Q51: How does meniscal repair compare to no repair in terms of osteoarthritis progression following ACL reconstruction?

A51: Current studies have not demonstrated long-term outcomes or OA progression favoring meniscus repair versus no repair, with most studies having only 2- to 3-year follow-up.

Q52: What factors should be considered when deciding between meniscal repair and resection during ACL reconstruction?

A52: Patient factors, such as age, BMI, and activity level may be important considerations that affect the value of meniscal preservation.

Q53: How does MCL treatment affect ACL graft survival in combined ACL/MCL injuries?

A53: Some evidence suggests that surgical treatment of the MCL in combined ACL/MCL injuries may decrease the risk of subsequent ACL graft failure compared to nonoperative MCL treatment.

Q54: Are there any short-term drawbacks to surgical treatment of the MCL in combined ACL/MCL injuries?

A54: Surgical treatment of the MCL may result in an early delay in return of quad strength and range of motion, but this typically normalizes by 2 years postoperation.

Q55: Has prophylactic knee bracing been shown to significantly reduce ACL injury rates in any studies?

A55: While some studies have shown trends toward fewer ACL injuries with bracing, the differences have not been statistically significant in available research.

Q56: What are some potential negative effects of widespread prophylactic knee bracing use?

A56: Widespread use could add significantly to the cost of sports participation, potentially heightening socio-economic bias, without clear evidence of injury reduction benefit.

Q57: Besides hop testing, what other criteria have been studied for determining readiness to return to sport after ACL reconstruction?

A57: Other criteria studied include muscle function tests, timing from surgery, measures of kinesiophobia, and various rating scales, but evidence for these is limited.

Q58: Is there evidence supporting the use of temporal criteria (e.g., 9 or 12 months postsurgery) for return to sport decisions?

A58: Evidence is mixed, with one study suggesting using 9 months as a criterion results in a lower rate of graft failure, while another found no difference when 12-month criteria are applied.

Q59: Do functional knee braces provide any measurable biomechanical benefits after ACL reconstruction?

A59: While functional braces may alter lower extremity biomechanics, the studies reviewed did not demonstrate any clinical benefit from these alterations in patients after isolated primary ACL reconstruction.

Q60: How might the use of functional bracing affect thigh muscle development after ACL reconstruction?

A60: Early bracing has been linked to decreased thigh circumference, which could potentially impact muscle strength and function.

Q61: Why is it important to assess the ability to bear weight when evaluating a potential ACL injury?

A61: The ability to bear weight can provide information about the severity of the injury and help differentiate between various knee injuries.

Q62: What is the significance of assessing varus and valgus laxity during the physical exam for ACL injury?

A62: Assessing varus and valgus laxity helps evaluate the integrity of the collateral ligaments, which can be injured in conjunction with the ACL.

Q63: How does the timing of ACL reconstruction affect the likelihood of meniscal tears?

A63: Several studies have shown that early ACL reconstructions had less meniscus injury than late ACL reconstructions at the time of surgery.

Q64: Is there a specific timeframe, after which the risk of additional injuries significantly increases?

A64: Some studies have shown that a delay of greater than 3 months was a predictor of more severe meniscus injury.

Q65: Are there any differences in knee stability between single- and double-bundle ACL reconstruction?

A65: While some individual studies have reported improved stability with double-bundle reconstruction, meta-analyses have not shown consistent, significant differences in stability outcomes.

Q66: How do long-term outcomes compare between single- and double-bundle ACL reconstruction?

A66: Current evidence shows no significant differences in long-term outcomes between single- and double-bundle ACL reconstruction.

Q67: How does the choice between autograft and allograft affect surgical time?

A67: Allograft use typically results in shorter surgical times compared to autograft, as there is no need for graft harvest.

Q68: Are there differences in the immune response between autograft and allograft ACL reconstructions?

A68: Allografts may elicit a greater immune response compared to autografts, which could potentially affect the graft incorporation process.

Q69: How does the choice between BTB and hamstring autografts affect donor site morbidity?

A69: Some studies have shown that BTB autografts are associated with greater donor site morbidity compared to hamstring autografts.

Q70: Are there differences in graft strength between BTB and hamstring autografts?

A70: Evidence recently shows a variable size of hamstring grafts, downgrading the level of evidence for BTB over hamstring to "moderate".

Q71: What are the potential harms associated with ACL training programs?

A71: Multiple studies have shown the benefit of exercise training in ACL injury prevention, while very few have demonstrated any harm.

Q72: Are ACL injury prevention training programs cost-effective?

A72: While the cost-effectiveness is not well studied, the potential savings from preventing ACL injuries likely outweigh the costs of implementing these programs.

Q73: What is the significance or importance of ALL reconstruction or LET in a hamstring autograft reconstruction?

A73: ACL injury incidence continues to increase, and the medical/financial/psychosocial impact of revision surgery is significant; thus, the evaluation of risk factors for reoperation is important.

Q74: Are there any specific patient groups who might benefit most from the addition of ALL reconstruction or LET?

A74: More research is needed to define specific indications, notably investigating the impact of particularly affected demographics like female adolescents, who experience the highest risk of graft failure.

Q75: What are some areas of further research regarding ACL repair versus reconstruction?

A75: Future research should look to identify ways of limiting revision surgery, involving topics such as patient selection based on tear location, biologic intervention, and surgical technique.

Q76: Has the standard of care for ACL repair for reconstruction changed?

A76: ACL reconstruction currently remains the standard of care for primary ACL injury.

Q77: Is aspiration of the knee an acceptable treatment for acute ACL tears?

A77: If evidence shows positive clinical implications, knee aspiration would be an acceptable treatment for acute ACL tears.

Q78: What further research needs to be conducted regarding the evidence for knee aspiration?

A78: Prospective, randomized trials are necessary to determine the effect of aspiration of the hematoma.

Q79: How does ACL reconstruction affect the risk of early-onset osteoarthritis compared to nonoperative treatment?

A79: Evidence is mixed, with some studies showing higher rates of osteoarthritis progression with ACL reconstruction, while others show no difference or favor reconstruction.

Q80: What factors should be considered when deciding between operative and nonoperative treatment for ACL tears?

A80: Factors include age, activity level, presence of associated injuries, degree of instability, patient goals, and willingness to modify activities.

Methodology

Q81: What happens during the review period for the ACL clinical practice guideline?

A81: The CPG draft undergoes a 3-week review period for additional input from external content experts, who provide written comments on a structured review form.

Q82: Who is solicited for nominations of individual reviewers for the guideline review?

A82: Specialty societies relevant to the topic are solicited for nominations of individual reviewers approximately 6 weeks before the final meeting.

Q83: What is the role of the chairs of the guideline work group during the review period?

A83: The chairs of the guideline work group review the draft of the guideline prior to dissemination.

Q84: How do some specialty societies participate in the review process?

A84: Some specialty societies ask their evidence-based practice (EBP) committee to review the guideline and consolidate their comments into one form.

Q85: What does the AAOS require from reviewers during the review period?

A85: Reviewers are required to disclose their conflicts of interest and provide their comments assembled into a single response form.

Q86: What happens to the comments received during the review period?

A86: The comments are reviewed by the chairs of the guideline work group, the manager of the AAOS CQV unit, and the Director of AAOS CQV, and initial responses to comments addressing methodology are drafted.

Q87: Who reviews the initial responses to comments on clinical practice and techniques?

A87: The initial responses to comments are reviewed by the chair and co-chair of the guideline work group, who respond to questions concerning clinical practice and techniques.

Q88: How are final revisions to the guideline handled?

A88: Final revisions are summarized in a report provided alongside the guideline document throughout the remainder of the approval processes and final publication.

Q89: How does the AAOS demonstrate responsiveness to input received during the review process?

A89: Following final approval of the guideline, all individual responses are posted on the AAOS website with a point-by-point reply to each noneditorial comment.

Q90: What are the approval steps for the final clinical practice guideline draft?

A90: The final CPG draft must be approved by the AAOS Committee on Evidence-Based Quality and Value, the AAOS Research and Quality Council, and the AAOS Board of Directors.

Summary of Options

Q91: What is the guideline's recommendation on the aspiration of the knee after injury?

A91: In the absence of reliable evidence, the guideline suggests that physicians may consider aspirating painful, tense effusions after knee injury based on clinical opinion

Q92: What does the guideline say about the option of ACL surgical reconstruction?

A92: The guideline states that ACL reconstruction can be considered to lower the risk of future meniscus pathology or procedures, particularly in younger and/or more active patients, and may improve long-term pain and function.

Q93: What is the recommendation for meniscal repair in conjunction with ACL tears?

A93: The guideline recommends considering meniscal preservation in patients with ACL tears to optimize joint health and function.

Q94: How should combined ACL/MCL tears be treated according to the guideline?

A94: The guideline suggests that nonoperative treatment of the MCL injury results in good patient outcomes, although operative treatment of the MCL may be considered in select cases.

Q95: What does the guideline recommend regarding prophylactic knee bracing?

A95: The guideline recommends against the routine use of prophylactic knee bracing as a preferred option to prevent ACL injury.

Q96: What is suggested about the return to sport after ACL reconstruction?

A96: The guideline suggests that functional evaluation, such as the hop test, may be considered as one factor to determine return to sport after ACL reconstruction.

Q97: What is the guideline's position on return to activity functional bracing?

A97: The guideline states that functional knee braces are not recommended for routine use in patients who have received isolated primary ACL reconstruction, as they confer no clinical benefit.

Q98: What are the guidelines for managing aspiration of the knee?

A98: The guideline suggests that physicians may consider aspirating painful, tense effusions after knee

injury, although this recommendation is based on clinical opinion due to the lack of reliable evidence.

Q99: What does the guideline mention about ACL reconstruction in terms of long-term outcomes?

A99: The guideline notes that ACL reconstruction may be considered to improve long-term pain and function, particularly in younger and/or more active patients.

Q100: What options are provided for managing meniscal tears in patients with ACL injuries?

A100: The guideline recommends meniscal preservation to optimize joint health and function in patients with ACL injuries.